

EduPro Full Project Report

Title

EduPro Student Segmentation and Personalized Course Recommendation System

Abstract

This project develops a learner-centric recommendation system using machine learning clustering techniques to identify learner segments and provide personalized course recommendations.

Introduction

Online learners have diverse goals, interests, and learning behaviors. Generic recommendations often fail to engage learners effectively.

Problem Statement

EduPro lacks a structured learner segmentation framework and personalized recommendation engine.

Objectives

Identify learner types, perform segmentation, generate personalized recommendations, and improve engagement.

Dataset Description

Users, Courses, and Transactions datasets are integrated to build learner profiles.

Feature Engineering

Features include total enrollments, spending, diversity score, preferred category, course ratings, and learning depth.

Methodology

Data preprocessing, feature engineering, clustering using K-Means, cluster validation using Silhouette Score, and recommendation generation.

Algorithms Used

K-Means Clustering, Hierarchical Clustering, StandardScaler, Label Encoding, and Content-Based Recommendation.

Exploratory Data Analysis

Analyze course enrollments, spending patterns, learner preferences, and category popularity.

Streamlit Dashboard

Interactive dashboard with learner explorer, cluster visualization, recommendation panel, and filters.

Results

Learners are grouped into meaningful segments such as Explorers, Specialists, Career Builders, and Casual Learners.

Benefits

Improved learner engagement, higher course completion rates, and increased platform retention.

Future Scope

Integration with deep learning recommendations, real-time analytics, and adaptive learning paths.

Conclusion

The project transforms EduPro from a generic recommendation platform into a personalized learning ecosystem.

Sample Streamlit Project Structure

app.py
Users.csv
Courses.csv
Transactions.csv
requirements.txt
Project_Report.pdf